

IBM Machine Learning Professional Certificate
Exploratory Data Analysis for Machine Learning - Course Project

A Project Report

on

**Exploratory Data Analysis and Hypothesis
Testing of The Titanic Survival Dataset**

by

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June, 2025

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1. Introduction and Summary of the Dataset

1.1 About the Project

In this project, the aim was to perform a comprehensive exploratory data analysis on a given dataset, which includes multiple steps including data cleaning, visualizations, feature encoding and engineering, hypothesis testing etc. and use the findings of these analyses to establish key takeaways and conclusions.

Furthermore, the analysis was used to formulate possible relations between different features and the target variable (if any), which can potentially be used to build a predictive linear regression model on the dataset.

1.2 Information about the Dataset

For this project, the dataset chosen was the Titanic Survival Dataset, available on Kaggle. This dataset lists the details of passengers traveling aboard the ocean liner RMS Titanic on its maiden voyage, as well as information on whether or not they survived the sinking of the ship on 15th April, 1912.

The information available include personal details such as the name, age, gender etc. of the passengers, as well as passenger details such as the passenger class, cabin, fare etc.

The dataset also includes a column specifying whether each passenger survived or not, and this feature served as the target variable in this project. The analysis aimed to predict if the passenger survival was purely random and based on luck, or whether certain demographics had a higher chance of survival.

The dataset was imported from the following URL:

<https://www.kaggle.com/datasets/yasserh/titanic-dataset/data>

1.3 Additional Information

This analysis was performed on the web-based Jupyter Notebook IDE, using the Python (Pyodide) kernel.

The coding was performed in Python 3.13.3, and employed the use of the following libraries for various steps and processes:

- Pandas
- NumPy
- Matplotlib
- Seaborn
- SciPy

1.4 Dataset Summary

The dataset spans 891 rows, each representing a single passenger, and 12 columns, each representing a certain detail about the passenger. It consists of 5 categorical (object) and 7 numerical (int, float) columns, as follows:

```
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   PassengerId     891 non-null    int64
1   Survived        891 non-null    int64
2   Pclass         891 non-null    int64
3   Name           891 non-null    object
4   Sex            891 non-null    object
5   Age           714 non-null    float64
6   SibSp         891 non-null    int64
7   Parch         891 non-null    int64
8   Ticket        891 non-null    object
9   Fare          891 non-null    float64
10  Cabin         204 non-null    object
11  Embarked      889 non-null    object
dtypes: float64(2), int64(5), object(5)
```

The numeric columns have the following data distribution:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

2. Data Cleaning and Prepping

2.1 Handling Null Values

The first step of the EDA process was to clean the raw, unfiltered data and ensure that the dataset is complete, consistent and usable in further data analysis. The cleaning processes included handling the missing values and outliers, and dropping columns that were not useful in further analysis.

Null values were present in three columns as follows:

Cabin	687
Age	177
Embarked	2

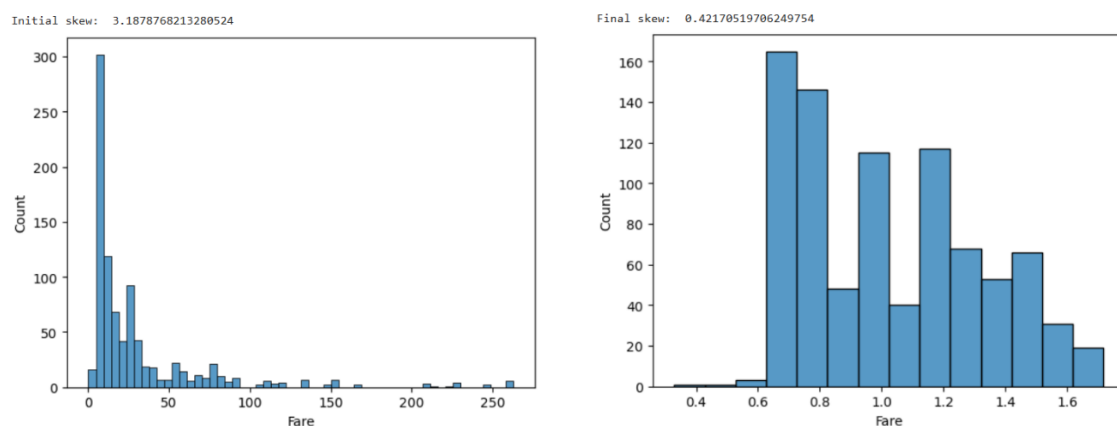
These were handled using methods appropriate to the feature;

- Null values in the 'Cabin' column were filled with the value 'Unassigned'.
- Null values in the 'Age' column were filled with the imputed value of column median, 28.00.
- Null values in the 'Embarked' column were filled with value of column mode, 'S'.

2.1 Handling Outliers and Skewed Features

Upon investigating the numeric columns, the 'Fare' column displayed significant outliers as well as a heavily skewed distribution. This was handled by dropping outlying fares at the higher end with value>300.

Furthermore, log transformation was performed on the columns to achieve a more normalized distribution, which helped to better visualize the spread of data. The initial and final distribution along with their respective skews were as follows:



3. Feature Engineering and Encoding

The next step was to create new features using the existing features of the dataset. The purpose of this step was to simplify certain features, and attempt to glean more information from categorical columns to use in our analysis.

This step was achieved using three processes, as follows:

- A new feature; 'Family_Size' was created by adding the 'SibSp' and 'Parch' features as separately analyzing the two groups did not yield anything fruitful.
- The column 'Sex' was encoded using ordinal encoding, and a new column 'Sex_Num' was created by mapping the 'male' and 'female' values to 1 and 2 respectively, allowing this feature to be included in analysis and visualizations.
- The column 'Embarked' was encoded using one-hot encoding, being replaced by three columns for each unique value; 'S','C','Q'. This allowed the columns to be used in numerical analyses.

Lastly, some unhelpful columns were dropped such as the 'Ticket' column, the values of which were an alphanumerical soup useless in the analysis, and the entire dataset was reordered to create a more sensible ordering of the features, as shown:

PassengerId		Name	Age	Sex	Sex_Num	Family_Size	SibSp	Parch	Pclass	Cabin	Fare	Src_S	Src_C	Src_Q	Survived
0	1	Braund, Mr. Owen Harris	22.0	male	1	1	1	0	3	Unassigned	7.2500	1	0	0	0
1	2	Cumings, Mrs. John Bradley (Florence Briggs Th...	38.0	female	2	1	1	0	1	C85	71.2833	0	1	0	1
2	3	Heikkinen, Miss. Laina	26.0	female	2	0	0	0	3	Unassigned	7.9250	1	0	0	1
3	4	Futrelle, Mrs. Jacques Heath (Lily May Peel)	35.0	female	2	1	1	0	1	C123	53.1000	1	0	0	1
4	5	Allen, Mr. William Henry	35.0	male	1	0	0	0	3	Unassigned	8.0500	1	0	0	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
886	887	Montvila, Rev. Juozas	27.0	male	1	0	0	0	2	Unassigned	13.0000	1	0	0	0
887	888	Graham, Miss. Margaret Edith	19.0	female	2	0	0	0	1	B42	30.0000	1	0	0	1
888	889	Johnston, Miss. Catherine Helen "Carrie"	28.0	female	2	3	1	2	3	Unassigned	23.4500	1	0	0	0
889	890	Behr, Mr. Karl Howell	26.0	male	1	0	0	0	1	C148	30.0000	0	1	0	1
890	891	Dooley, Mr. Patrick	32.0	male	1	0	0	0	3	Unassigned	7.7500	0	0	1	0

891 rows × 15 columns

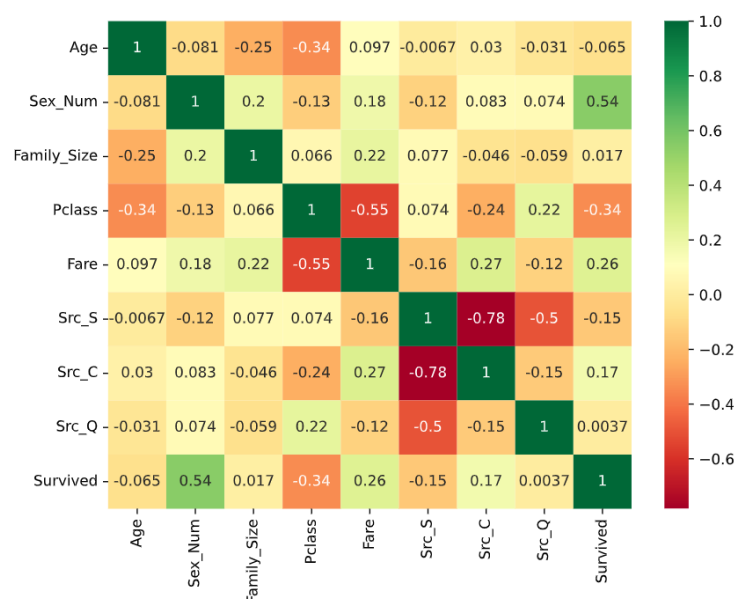
4. Data Visualizations

The next section of the analysis involved creating various visualizations on the entirety, as well as on partial sections, of the dataset. The main objective of this step was to investigate relations between different features, and how certain features affected the target variable.

A wide range of visual diagrams were used, mainly from the Seaborn library, in combination with functionalities provided by matplotlib and pandas.

4.1 Feature Correlation Heatmap

The `.heatmap()` function of Seaborn was used to create a heatmap of the correlation matrix between the major numerical columns of the dataset.

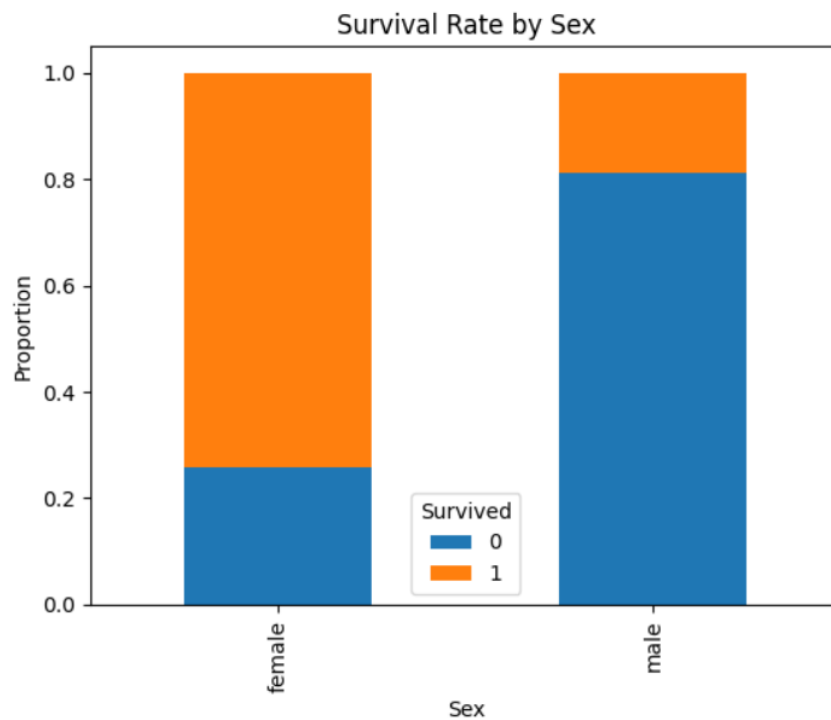


- Key observations from this visualization are as follows:
 - High inverse correlation between 'Pclass' and 'Survived', indicating the higher survival odds of upper-class passengers.
 - High positive correlation between 'Sex_Num' and 'Survived', indicating the higher survival rates of female passengers.
 - Unusual medium-high positive correlation between 'Fare' and 'Src_C', indicating that a majority of upper-class passengers paying the higher fare probably embarked from Cherbourg, France.

4.2 Exploring Specific Feature Correlations with Target

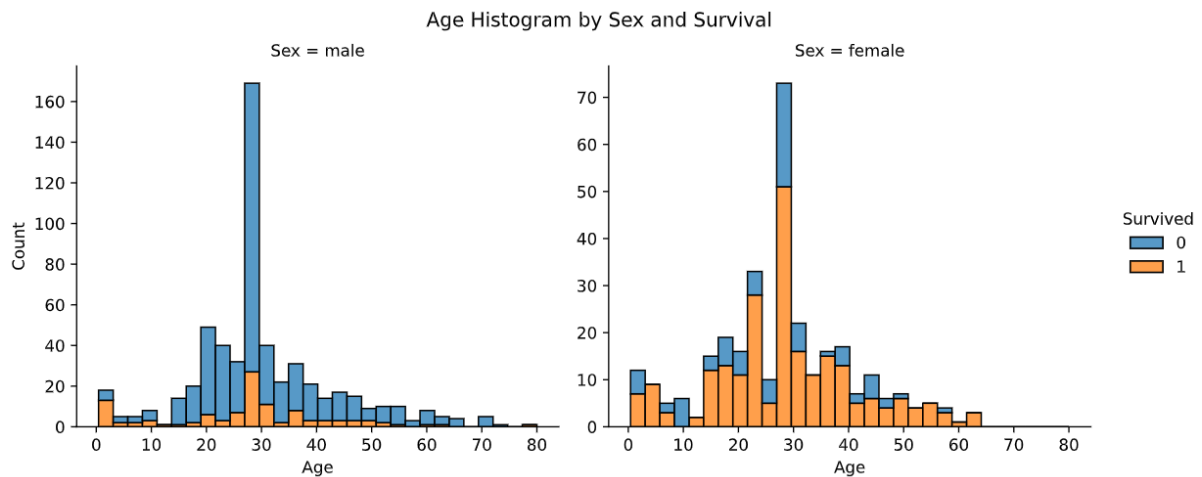
This section visualized the relationship of specific variables with the target variable to determine if certain demographics had a higher survival rate, and if so, how skewed was the distribution.

The correlation between gender and survival was more clearly visualized using a normalized stacked bar graph, created with the `.plot()` function of Matplotlib.



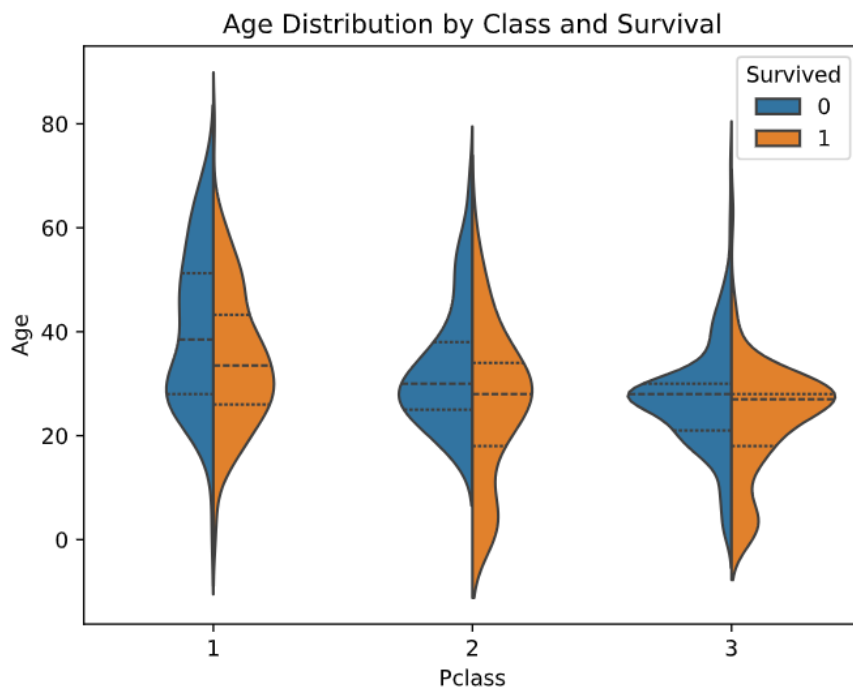
- Key observation was the stark difference between the percentage survival rates of men and woman, as inferred from the earlier correlation.
- A follow-up that naturally arose was that, since lifeboat priority was given to 'woman and *children*, was this distribution even more skewed across different age groups?

The aforementioned query was visualized by using a stacked faceted histogram created using the `.distplot()` function of Seaborn.



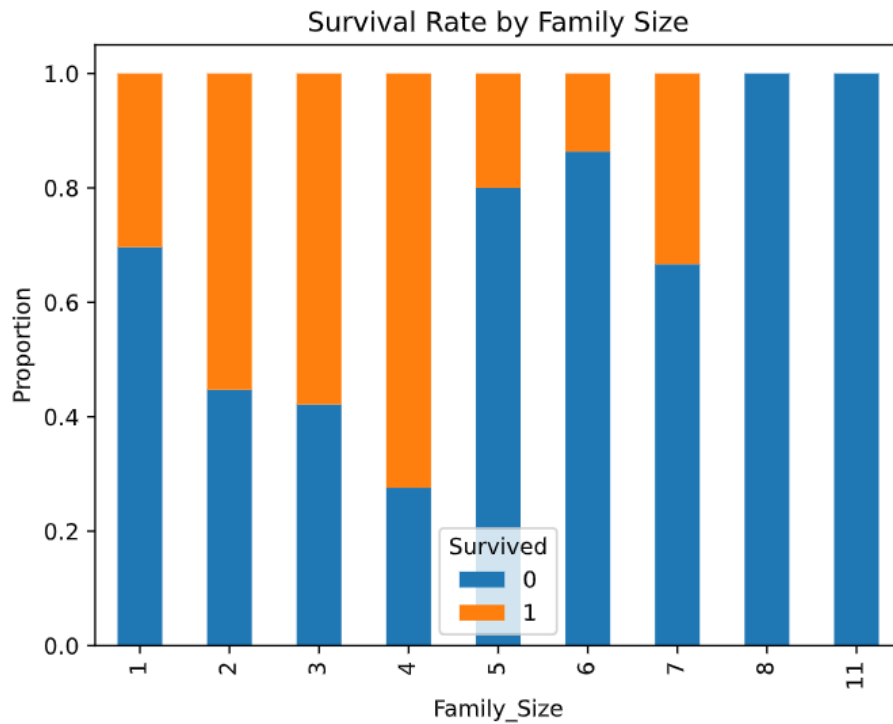
- The visualization suggested a possible effect of age groups on the survival rates. This insight was tested as a hypothesis in a later step of the analysis.

The distribution of survival rates across different ages was further visualized in the context of different passenger classes using the `.violinplot()` function of Seaborn.



- While this visualization did suggest higher survival percentage of the upper-class as a whole, it refuted the claim that passenger class affected the survival rates of younger children, who had very high survival rates in the middle and lower classes as well.

Lastly, the `.plot()` function of Seaborn was used to create another normalized stacked bar graph, this time to visualize the correlation between the 'Family_Size' and 'Survived' columns.



- Key observations from this visualization were the highest survival rate of medium sized families of 4 members, and drastically lower survival of larger families.
- Furthermore, survival rates were lower for passengers traveling alone.

5. Hypothesis Testing

The final step in the analysis was the formulation of multiple hypotheses based on the earlier findings and visualizations. These hypotheses were primarily aimed at exploring what features, or combination of them, could be best used to predict the target variable, 'Survived'.

5.1 Survival by Passenger Class

- ❖ **Null Hypothesis (H_0)** – Upper-class passengers ('Pclass' = 1) had the same survival rate as middle-class ('Pclass' = 2) and lower-class ('Pclass' = 3) passengers.

Alternate Hypothesis (H_a) – Upper-class passengers ('Pclass' = 1) had a higher survival rate than middle-class ('Pclass' = 2) and lower-class ('Pclass' = 3) passengers.

- *Rationale* – The wealthier upper-class passengers might have been given priority while boarding lifeboats, and thus, had a higher survival rate.
- *Evidence* – The violin-plot suggested higher percentage survival rate of upper-class passengers.

5.1 Survival by Family Size

- ❖ **Null Hypothesis (H_0)** – Medium-sized families had the same survival rate as larger families.

Alternate Hypothesis (H_a) – Medium-sized families of 2-4 members had a higher survival rate than families larger than 4 members.

- *Rationale* – The logistical issues involved in rescuing all members of large families might have led to lower survival rates for such passengers.
- *Evidence* – The normalized stacked bar graph suggested lower survival rates for large families of 5 or more members.

5.1 Survival of Specific Age Group based on Gender

- ❖ **Null Hypothesis (H_0)** – Female passengers of the dependent age group had the same survival rate as all the other groups of passengers.

Alternate Hypothesis (H_a) – Female passengers of the dependent age group ('Age' ≤ 16 | 'Age' ≥ 50) had the highest survival rate than the other groups of passengers.

- *Rationale* – The priority of 'women and children first' offered while boarding the lifeboats might have resulted in the highest survival of dependent age group of women.
- *Evidence* – The faceted stacked histogram suggested the highest rates of survival in the youngest and oldest age groups of female passengers.

5.4 Significance Test for Hypothesis 3

Out of the aforementioned hypotheses, further analysis was performed on the third hypothesis, using a significance test.

The test was carried out in the following manner:

1. First, a separate column was created named 'Demographic' and the entries were grouped under one of the following four categories (Working indicates $16 < \text{'Age'} < 50$, Dependent indicates otherwise):

- a. Male Working (MW)
- b. Male Dependent (MD)
- c. Female Working (FW)
- d. Female Dependent (FD)

2. A Chi-Squared (χ^2) contingency test was performed on the columns 'Demographic' and 'Survived' to ensure that associativity was present and the different survival rates weren't random.

This was carried out using the `.chi2_contingency()` function of the `scipy.stats` library.

The test yielded an extremely low p-value of $1.016e-57$, indicating very high associativity and suggesting that passenger demographic strongly influenced survival.

3. Finally, the 'FD' group was subjected to pairwise one-tailed two-proportion z-tests with all the other groups, keeping a Bonferroni-corrected alpha value of $0.05 / 3 = 0.0167$.

This was performed using the `proportions_ztest()` function.

The results were as below:

```
FD > FW?  z =  0.10   p = 0.4613
Fail to reject null hypothesis for:  FD  v/s  FW
FD > MD?  z =  6.04   p = 0.0000
Reject null hypothesis for:  FD  v/s  MD
FD > MW?  z = 10.57   p = 0.0000
Reject null hypothesis for:  FD  v/s  MW
```

4. As seen above, the test failed to reject the null hypothesis for the pairwise comparison of the female dependent and female working demographics, but rejected the null hypothesis for the other two comparisons.

6. Key Findings and Insights

The progressive analysis of this dataset revealed some key insights about the features and how they relate to the target column. Each step of the EDA process proved critical in understanding the correlations between the features.

The visualizations revealed that features such as 'Sex_Num', 'Family_Size', 'Pclass', 'Fare', and to some extent even 'Age', all influenced the survival rates of passengers in some capacity. Passengers of the following demographics had significantly higher survival;

- 'Sex_Num' = 2 i.e. female passengers.
- $2 \leq \text{'Family_Size'} \leq 4$ i.e. passengers with medium-sized families of 2-4 members.
- 'Pclass' = 1 i.e. upper-class passengers.

Further inferences from the visualizations included the following:

- Younger passengers had the highest survival rates of any age group, regardless of their gender or passenger class.
- Passengers embarking from Src_C i.e. Cherbourg, France, primarily belonged to upper-class, and by extension, had a higher rate of survival than passengers from the other two ports.

Finally, the insight gained from performing hypothesis testing on different demographic classes defined by a combination of 'Age' and 'Sex' revealed that there was no major difference in the survival rates of working age group and dependent age group but both were significantly higher than their male counterparts.

7. Conclusion

In conclusion, this exploratory data analysis proved to be a comprehensive and insightful method to understand and explore the different aspects of this dataset and gauge the relationships that existed between its features.

The Titanic Survival dataset was chosen, and a step-by-step process was followed to analyze it. First, a general health check was carried out on the dataset to understand it, followed by cleaning and prepping the data, then encoding some new features and dropping redundant ones, after which the data was visualized through various graphs and plots, and its insights used to formulate multiple hypotheses and rigorously test one of them. In the end, the key takeaways were summarized and concisely presented in the previous section.

Now that the dataset has been thoroughly analyzed, the possible next steps include:

- Cleaning up and polishing the code by making it concise and commenting it to make it understandable and readable for any further use.
- Train a linear regression model on this dataset, and test it with the Titanic test dataset on Kaggle.
- Work on a possible pipeline prototype if such a model were to be deployed in practice.

These will ensure that the takeaways from this EDA are optimally used and are beneficial in practical applications further down the line.